

ACTIVITY 5 — FACILITATOR GUIDE

DMAIC Overview, Problem Statement, Scope, and Stakeholders

TYPE	DURATION	DMAIC PHASE	MAPS TO
Core Lab	45 min	DEFINE	A1, A2

Purpose of this activity

Support the Define phase of a DMAIC project (A1, A2).

Step up from PDCA to the full DMAIC roadmap. Clarify what each phase delivers, where a Yellow Belt contributes most, and prepare defensible Define-phase information for the project leader.

What learners produce

A DMAIC phase table, a refined problem statement, a stakeholder map and a benefit estimate in business terms.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- A laptop with Excel, Google Sheets or LibreOffice Calc
- The CSV files in this activity's data/ and templates/ folders
- Flipchart or A3 sheet and markers
- The course slides for the scenario and the framework

Set-up

Groups of 3-5. Distribute one worksheet per learner and one checklist per group. Every activity runs on the same continuous Contoso Service Desk scenario, so remind learners where this one sits in that story before starting.

Scenario / briefing material

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder

templates/ — the worksheets you complete:

- dmaic-phase-table.csv (phase, purpose, key_deliverable, yellow_belt_support_role)
- stakeholder-map.csv (stakeholder, interest, influence_H_M_L, interest_H_M_L, engagement_approach)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Summarise all five DMAIC phases: purpose, key deliverable and Yellow Belt support role.	Learners jump to Improve — hold them in Define until the problem is defensible.
2	Refine the problem statement so it is specific, measurable and solution-free.	Benefit in business terms means money, time or risk, not 'better quality'.
3	Identify stakeholders and classify each by influence and interest.	
4	Define the project scope and state the expected benefit in business terms.	
5	Link the problem back to the CTQ requirements captured in Lab 2.	
6	Confirm which DMAIC phases a Yellow Belt can support most strongly.	

Suggested timing

Minutes	Phase
10 min	Summarise the five DMAIC phases
10 min	Refine the problem statement
10 min	Map stakeholders by influence and interest
10 min	Define scope and benefit
5 min	Link back to the Lab 2 CTQs

Total: 45 minutes

Discussion & decision prompts

1. Which DMAIC phase does a Yellow Belt support most strongly, and why?
2. Which stakeholder could stop this project, and how will you engage them?
3. How does the problem connect back to the CTQs you captured in Lab 2?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what the data showed.
- Link the activity explicitly back to the DEFINE phase of DMAIC.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- Who are the high-power, low-interest stakeholders in your own work?
- What business benefit would make your manager fund an improvement?

Facilitator notes — what to watch for

- Learners jump to Improve — hold them in Define until the problem is defensible.
- Benefit in business terms means money, time or risk, not 'better quality'.

Success criterion

Your problem statement passes the 'no solution named' test and every stakeholder has a defined engagement approach.

Assessment link

This activity builds the DEFINE phase of DMAIC. All activities are in-class activities assessed indirectly through the open-book Case Study (CS), which reuses the same techniques on the same Contoso Service Desk scenario and the same figures (400 tickets, yield 76%, DPMO 50,000). Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.